

Matthew Lau

Cupertino, CA | (408) 596-1554 | <https://matthew-coding-portfolio.vercel.app/> | mlau191@cs.washington.edu | <https://github.com/mattycakes14>

EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science, Paul G. Allen School

Graduation: Jun 2028

GPA: 3.71 /4.0 | Dean's List

Coursework: Object-Oriented Programming, Data Structures and Parallelism, Hardware/Software Interface, Linear Algebra, Multivariable Calculus, Probability & Computational Complexity, Machine Learning, Systems Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL, Go, C, C++

Frameworks & Libraries: React, React Native, Next.js, Flask, FastAPI, Node.js, Express.js, Tailwind, LangChain, NumPy, TensorFlow, PyTorch

Databases & Tools: Firebase, Supabase, Redis, Qdrant, Pinecone, Docker, Gunicorn, SQLAlchemy, Alembic, AWS (EC2, Auto Scaling Groups, EKS, Lambdas, CloudWatch), Kubernetes, Terraform, Git, GitHub Actions

EXPERIENCE

Amazon

Palo Alto, CA

Incoming Software Engineer Intern – Distributed Databases (Aurora)

Sep 2026 – Dec 2026

Workday

Pleasanton, CA

Software Engineer Intern – Platform/DevOps

Jun 2026 – Sep 2026

- Diagnosed why a fleet-wide OpenTelemetry disable config was silently ignored, found 20 of 34 services still actively exporting, and guarded the export path – eliminating **~632k daily error log spam** and unlocking decommission of the deprecated collector infrastructure (ALB, target groups, ASGs)
- Migrated **60 AWS Lambda functions** from x86-64 to arm64 (Graviton), auditing Docker image layers to verify every binary and transitive dependency resolved to arm64 before cutover; **~20% lower compute cost per GB-second**
- Migrated **8+ repositories** from hardcoded provider configurations to a reusable Terraform module, eliminating duplicated definitions across repos and centralizing how services retrieve secrets from AWS Secrets Manager
- Closed a gap where the monitoring stack (Grafana and Prometheus) was itself unmonitored, provisioning CloudWatch alarms in Terraform on ALB target-group health and routing them through SNS to PagerDuty under escalation policy
- **Top 5 Finalist** among 36 intern teams across the US, Canada, and Costa Rica in Workday's intern hackathon; designed an agentic AI career pilot for early-career talent and pitched it to the VP of Software Development and 100+ attendees

UW Medicine

Seattle, WA

Software Engineer – Full Stack

Feb 2026 – Present

- Added consistent success/error feedback to all write operations using Notistack across a Next.js codebase, surfacing async and API failures via try/catch and status-code checks with contextual messages
- Rewrote **Firestore** security rules and built Next.js API routes for role assignment, enforcing role-specific database permissions for admins, leads, and volunteers across **100+ service programs**

Clad Labs (YC F25)

San Francisco, CA

Software Engineer Intern – AI Tools

Jun 2025 – Sep 2025

- Architected an ad-content extraction pipeline (scraping + OCR) that increased ad-text coverage by **22%** and website-text by **33%**, generating semantic embeddings in Pinecone to match user prompts to relevant ads
- Built Flask API routes to recommend developer-tool advertisements inside AI coding assistant chats, using LangGraph for routing prompts and Supabase for storing user tool preferences for **10,000+ developer waitlist**

PROJECTS

Eat Together | *React Native, Firebase, JavaScript*

Jan 2025 – Sep 2025

- Built a React Native restaurant-discovery feature (preference quiz + swipe interface, Firebase-backed) with a Yelp API-driven scoring algorithm weighting dietary restrictions, cuisine, and price for **500+ users**